



TIME: 3 HRS

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief.

 $02 \times 7 = 14$

Q no.	Question	CO	Level
a.	How are objects implemented in java?	1	K4
b.	Differentiate between method overloading and method overriding in java.	1	K4
c.	What is the purpose of throw and throws keywords in java?	1	K4
d.	Explain how to change the font style and size in java graphics program.	1	K4
e.	What is the difference between multithreading and multitasking?	1	K4
f.	What is purpose of <i>JCheckBox</i> ?	3	K4,K5
g.	Define Encapsulation and Polymorphism.	5	K3,K4

SECTION B

2. Attempt any *three* of the following:

07 x 3 = 21

a.	Explain in details about variable in java.	1	K4
b.	What is object cloning in java? Differentiate between shallow and deep copy.	1	K4
c.	Write a java program to read content from one file and copy it to another file	2	K6
d.	What are daemon threads in java?	1	K4
e.	Write short note on Graphics 2D.	2	K6

SECTION C

3. Attempt any *one* part of the following:

07 x 1 = 07

a.	What is constructor and also explain how sub class constructor implicit calls super class constructor in java.	1	K4
b.	Explain the concepts of classes and objects in java with a simple example.	3	K4,K5

4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

a.	Explain the detailed about various type of inheritance in java with neat sketch.	5	K3,K4
b.	What is JVM? Explain the internal architecture of JVM.	3	K4,K5

5. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

a.	Explain the concept of exception handling in java. Describe how the throwing and catching mechanism works.	3	K4,K5
b.	Discuss the various type of file in java and explain the process of file handling.	2	K6

6. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

a.	Explain the concept of bounded types in java Generics.	3	K4,K5
b.	Explain the life cycle of thread in java with diagram.	2	K6

7. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

a.	What is event handling in java? Explain in detail.	2	K6
b.	What is java swing? How is it different from AWT? Explain in detail.	2	K6